

Chapter 2

Basic Concepts

The first step in any engineering analysis is the precise description of what is being studied. In thermodynamics, a **system** is defined as a *quantity of matter or a region in space chosen for study/analysis*. The shape or volume of the system being analyzed is not necessarily constant, e.g., when a gas in a cylinder is compressed by a piston or a balloon is inflated. Everything external to the system is considered to be a part of the system's **surroundings**. The system is distinguished from its surroundings by a specified **boundary**, which may be at rest or in motion.

Two basic kinds of systems are used for analysis – **closed systems** and **control volumes**. Closed system refers to a fixed quantity of matter (fixed mass), while a control volume is an identifiable region in space through which mass may flow. A closed system is also commonly referred to as **control mass**, and the term control volume is sometimes interchanged with **open system**. When the terms *control mass* and *control volumes* are used, the system boundary is often referred to as a *control surface*.

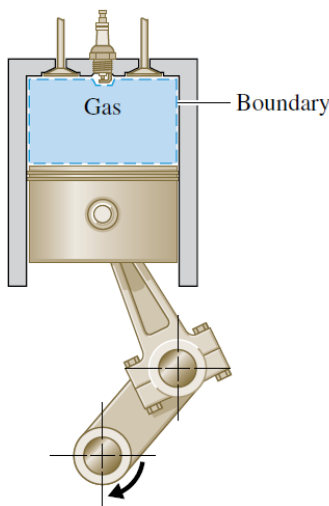


Figure 2.1: Example of a closed system: Gas enclosed in an IC Engine cylinder when valves are closed.

No mass can flow through the boundaries of a closed system. An example is shown in **Figure 2.1**. When the valves are closed the region enclosed by dashed rectangle represents a closed system. Here, as the piston moves, the volume of the closed system varies over time; the quantity of matter (total mass) of the system, however, remains constant.

In analysis of thermodynamic systems such as turbines, pumps, refrigeration systems etc. it is quite common to conduct the study on a control volume, e.g., see **Figure 2.2**. Control volumes are very commonly used in systems where fluid is flowing because it is not possible to keep track of the boundaries of a control mass. Analyses conducted using control volume are tractable especially when the study is experimental in nature. The measurements made using sensors/probes are useful in analyses employing control volumes whose boundaries align with geometrically identifiable features. Analyses based on control mass are used in few methods of theoretical analyses.

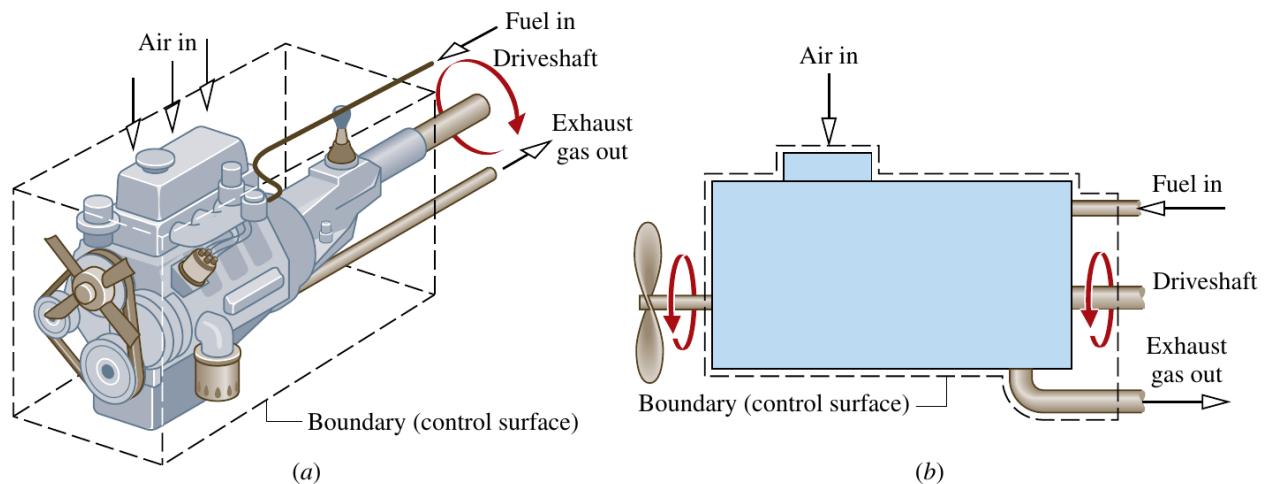


Figure 2.2: Example of a control volume (open system): An IC Engine with its fluid lines.

2.1 Molecular vs continuum

In thermodynamics, there are two approaches with which systems are analysed: *microscopic* and *macroscopic*. The **microscopic** approach of thermodynamics, known also as *statistical* thermodynamics, studies the structure of matter, e.g., molecular structure and intermolecular interactions. The matter is assumed to be made up of discrete molecules, with gaps in between them. The statistics, averages and moments over many molecules, are used to infer the macroscopic behaviour of a thermodynamic system. This approach is also called the *molecular approach*. Applications like lasers, plasma, high-speed gas flows, cryogenics make use of the statistical approach.

The **macroscopic** or *continuum* approach is concerned with the gross or overall behavior of collection of molecules at the macro level. This is also sometimes called *classical* thermodynamics. Here structure of molecules, atoms, or subatomic levels is not considered. The gross behavior is estimated/ modeled using observation of the overall system. The matter is assumed to be a continuum, which means there are no *gaps* in matter. Quantities like density, viscosity, velocity etc. are continuous functions of space.

2.2 Process, State & Process

A **property** is a macroscopic characteristic of a system such as mass, volume, energy, pressure, and temperature to which a numerical value can be assigned at a given time. The word **state** refers to the condition of a system as described by its properties. A state can be represented by a mathematical set consisting of system's properties as its entries/elements. Since there are normally relations among the properties of a system, the state often can be completely specified by providing the values of a subset of properties. All other properties can be determined in terms of these few.

When any of these property of a system changes, the state changes and the system is said to undergo a **process**. The series of states through which a system passes during a process is called the **path**. A process is a transformation from one state to another. A system is said to be at **steady-state** if none of its properties changes with time, i.e., time derivative of any property is zero. There are many quantities that are not properties, such as mass flow rates and energy transfers by work and heat.

A large number of engineering devices operate for long periods of time under the same conditions. Process involving such devices can be closely approximated by an idealized process, called the **steady-flow process**, which can be defined as a process during which fluid flows through a control volume steadily. That is, the fluid properties can change from point to point within the control volume, but at any fixed point they remain the same during the entire process.

Thermodynamic properties can be divided into two classes: extensive and intensive. A property is called **extensive** if its value for an overall system is the sum of its values for the parts into which the system is divided. Mass, volume, energy and several other properties introduced later are extensive. Extensive properties depend on the size or extent of a system. **Intensive** properties are not additive. Their values are independent of the size or extent of the system, and may vary spatially at any given instant of time. Intensive properties can be functions of both space and time, whereas extensive properties can vary only with time. Specific volume, pressure, and temperature are examples of intensive properties.

2.3 Equilibrium

In mechanics, equilibrium means balance of all forces and moments. In thermodynamics, balance of other influences are also taken into account, e.g. thermal, phase and chemical equilibrium. In **thermal equilibrium**, temperature is same throughout the system. In **phase equilibrium**, the mass of each phase reach equilibrium levels and stays there. In **chemical equilibrium** the chemical composition does not change with time, that is, no chemical reaction occurs. In an equilibrium state, there are no unbalanced potentials (or driving forces) within the system or at boundaries. A system in equilibrium experiences no changes when it is isolated from its surroundings.

When a process proceeds in such a manner that the system remains infinitesimally close to an equilibrium state at all times, it is called **quasi-static**, or **quasi-equilibrium** process. A quasi-equilibrium process can be viewed as a sufficiently slow process that allows the system to adjust internally to always remain infinitesimally close to equilibrium.

2.4 The State Postulate

The state of a system is fully described by its properties. However, as mentioned earlier, we do not need to specify all properties in order to fix a state. Once a sufficient number of properties are specified, the rest of the properties can automatically be calculated. That is, by specifying a certain number of properties, we can **fix a state**. A simple compressible system is completely specified by only two independent, intensive properties. For a single phase system, temperature and pressure are independent properties. However, for systems undergoing phase change, the pressure and temperature become dependent, and cannot be used to fully define a system.

2.5 Temperature

Informally, temperature has been thought of as a measure of “hotness” or “coldness”. Based on our physiological sensations, e.g., our sense of touch, we can express temperature only qualitatively using words like, freezing, cold, warm, hot, etc. We cannot assign numerical values to temperatures

based on physiological senses alone. Furthermore, our sense of touch may be misleading. A metallic object, for example, may feel much colder than a wooden one even when both are at the same temperature.

Several material properties change in a repeatable and predictable way with change in temperature. The commonly used mercury-in-glass thermometer is based on the expansion of mercury with temperature. It is a common experience that a cup of hot coffee left on the table eventually cool off and a cold drink eventually warms up. When two bodies at different temperatures are brought into contact, heat is transferred from object at higher temperature to the one at lower temperature until both bodies attain the same temperature. At that point, heat transfer stops and the two bodies are said to have attained **thermal equilibrium**. The equality of temperature is the only requirement for thermal equilibrium.

The **zeroth law of thermodynamics** states that if two bodies are in thermal equilibrium with the third body, they are in thermal equilibrium with one another. The fact may seem very basic, it cannot, however, be concluded from other laws of thermodynamics. It serves as a basis for validity of temperature measurement. By replacing the third body with a thermometer, the zeroth law can be restated as: *two bodies are in thermal equilibrium if both have the same temperature reading even if they are not in contact*.

The zeroth law of thermodynamics was first formulated by R. H. Fowler in 1931. Its value as a fundamental physical principle was recognized more than half a century after the formulation of the first and the second laws of thermodynamics.

2.6 Temperature Scales

Temperature scales enable us to use a common basis for temperature measurements, and several have been introduced through history. All temperature scales are based on some easily reproducible states such as freezing and boiling of water, which are also called the *ice point* and the *steam point*, respectively. The temperature scales used in SI and the English system today are the **Celsius scale**, and the **Fahrenheit scale** respectively. These are named after Swedish astronomer A. Celsius and German instrument maker G. Fahrenheit, respectively. On the Celsius scale, the ice and steam points were originally assigned values of 0 and 100° C, respectively. The corresponding points on the Fahrenheit scale are 32 and 212°F. These are often called *two-point scales* since temperature values are assigned at two different points.

In thermodynamics, it is very desirable to have a temperature that is independent of the properties of any substance. Such a temperature scale is called a **thermodynamic temperature scale**. The thermodynamic scale in the SI unit is the **Kelvin scale**, named after Lord Kelvin (1824-1907). The temperature unit on this scale is **kelvin**, which is designated by K, not °K. The lowest temperature on the Kelvin scale is absolute zero, or 0 K. The thermodynamic scale in the English system is the **Rankine scale**, named after William Rankine (1820-1872). The temperature unit on this scale is the **rankine**, which is designated by R. The temperature of 0 K corresponds to 0°R. One unit on the Rankine scale is equivalent to one Fahrenheit, and one unit on the Kelvin scale is equal to 1°C.

Any object with at least one measurable property that changes as its temperature changes can be used as a thermometer. Such a property is called a thermometric property. The particular substance that exhibits changes in the **thermometric property** is known as a *thermometric substance*. A familiar device for temperature measurement is the liquid-in-glass illustrated in **Figure 2.3(a)**. It consists of a glass capillary tube connected to a bulb filled with a liquid such as alcohol and sealed at the other end.

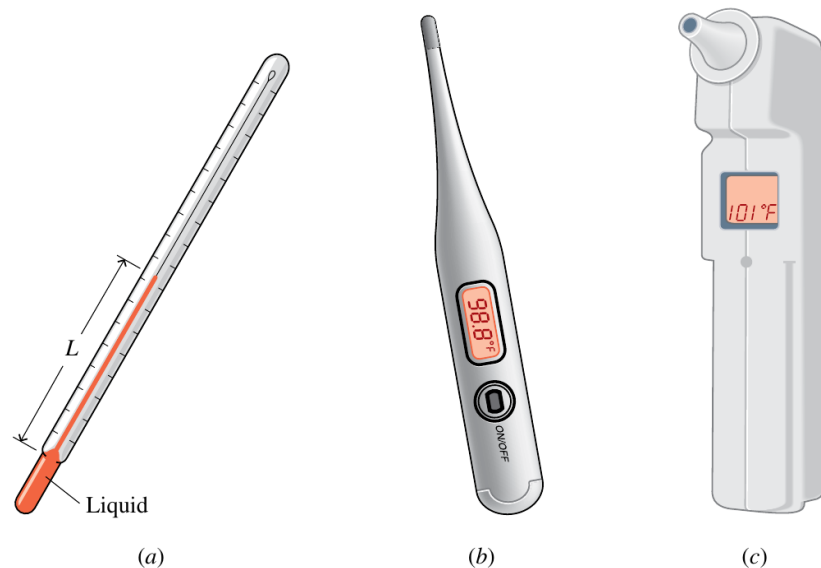


Figure 2.3: Thermometers: (a) Liquid-in-glass (b) Electrical-resistance (c) Infrared-sensing ear thermometer.

The space above the liquid is occupied by the vapor of the liquid or an inert gas. As the temperature increases, the liquid expands in volume and rises in the capillary. The length L of the liquid in the capillary depends on the temperature. Accordingly, the liquid is the thermometric substance and L is the thermometric property.

More accurate sensors known as **thermocouples** are based on the principle that when two dissimilar metals are joined, an electromotive force (emf) that is primarily a function of temperature will exist in a circuit. Thermocouples utilize pairs of metals such as platinum & alloy of platinum and rhodium; copper and constantan; iron and constantan etc. **Electrical-resistance** sensors are another important class of temperature measurement devices. These sensors are based on the fact that the electrical resistance of various materials changes in a predictable manner with temperature. The materials used for this purpose are platinum, nickel or copper or semiconductors. Devices using conductors are called *resistance temperature detectors*. Semiconductor types are called *thermistors*. An example is shown in **Figure 2.3(b)**.

A variety of instruments measure temperature by sensing radiation, such as the ear thermometer shown in **Figure 2.3(c)**. They are known by terms such as **radiation thermometers** or **optical pyrometers**. This type of thermometer differs from those previously considered because it is not required to come in contact with an object to determine its temperature, an advantage when dealing with moving objects or objects at extremely high temperatures.

2.7 Pressure

Pressure is defined as a normal force exerted by a fluid per unit area. The counterpart of pressure in solids is *normal stress*. The SI unit of pressure is newtons per square meter (N/m^2), which is also referred to as pascal (Pa). The actual pressure at a given position is called **absolute pressure**, and it is measured relative to absolute vacuum (i.e., absolute zero pressure). Most pressure-measuring devices, however, are calibrated to read zero in the atmosphere, and so they indicate the difference between absolute and local atmospheric pressure. This difference is called **gage pressure**, P_{gage} ,

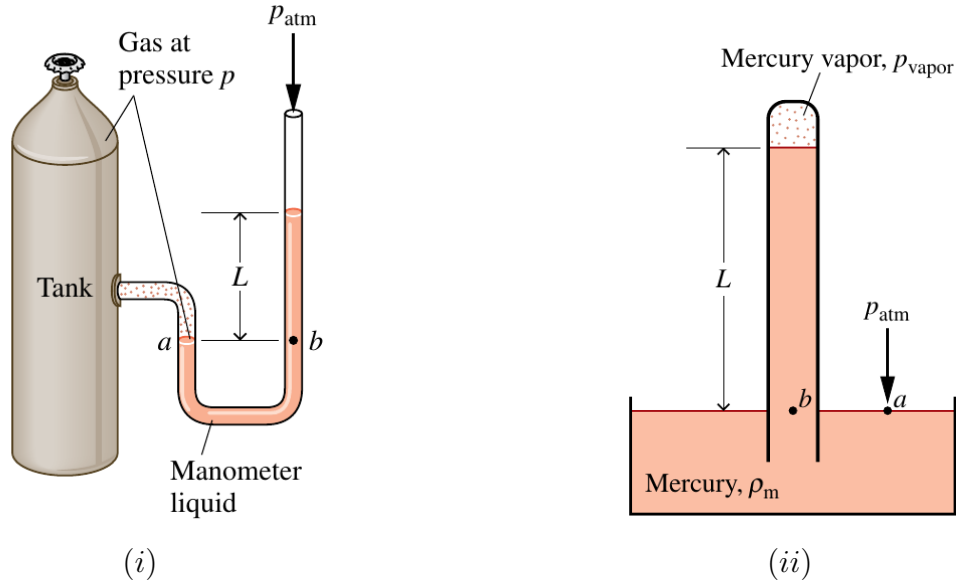


Figure 2.4: Pressure measuring devices: (i) Manometer (ii) Barometer.

which can be positive or negative. If P_{gage} is negative, the pressure is called **vacuum pressures**. The absolute, gage and vacuum pressures are related by

$$P_{\text{gage}} = P_{\text{abs}} - P_{\text{atm}} \quad (2.1)$$

$$P_{\text{vac}} = P_{\text{atm}} - P_{\text{abs}} \quad (2.2)$$

Manometers and **barometers** measure pressure in terms of the length of the column of liquid such as mercury, water or oil. The manometer shown in **Figure 2.4(i)** has one end open to atmosphere and the other attached to a tank containing a gas at uniform pressure. When the level of liquid in the manometer stabilizes, the pressures at equal elevations are equal, e.g., at points a and b .

$$p_a = p_b = p_{\text{atm}} + \rho g L \quad (2.3)$$

The barometer shown in **Figure 2.4(ii)** is formed by closed tube filled with liquid mercury and a small amount of mercury vapor inverted in an open container of liquid mercury. Since the pressures at points a and b are equal, a force balance gives the atmospheric pressure as:

$$p_{\text{atm}} = p_{\text{vapor}} + \rho_m g L \quad (2.4)$$

Here, p_{vapor} is pressure of mercury vapor which is much smaller than the atmospheric pressure. So **Equation (2.4)** can be approximated as $p_{\text{atm}} = \rho_m g L$. The pressures measured with manometers and barometers are frequently expressed in terms of the length L in millimeters of mercury (mmHg), inches of mercury (inHg), inches of water (inH₂O), and so on.

Pressure can be measured by other means as well. An important class of sensors utilizes the **piezoelectric effect**. A charge is generated within certain solid materials when they are deformed. This mechanical input/electrical output provides the basis for pressure measurement as well as displacement and force measurements. Another important type of sensor employs a diaphragm that deflects when a force is applied, altering an inductance, resistance, or capacitance. They are often paired with automatic data acquisition systems.

Bibliography

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